Regression Analysis

Complete Exam Answer Guide

10	5	R^2
Questions	OLS Properties	& Adj R^2

Based strictly on course materials: Testing of Hypothesis PPT (Chapter 10), 18.-Hypothesis-Testing.pdf, and uploaded course notes

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Q1 | 10 Marks | 2021-22 Final & Re-exam — REPEATED VERBATIM | REPEATED

Explain all the assumptions of Linear Regression. Mention the model of linear regression with all variables explained. Mention the properties of OLS estimators.

The Linear Regression Model

Linear regression models the relationship between a dependent variable Y and one or more independent variables X. The simple linear regression model is:

Y = beta_0 + beta_1 * X + epsilon

Y = dependent variable (response) | X = independent variable (predictor) beta_0 = intercept (value of Y when X = 0) | beta_1 = slope (change in Y per unit change in X) epsilon = error term / residual (captures all variation in Y not explained by X)

The OLS (Ordinary Least Squares) method estimates beta_0 and beta_1 by minimising the sum of squared residuals: minimise Sum(Yi - Y_hat_i)^2.

5 Assumptions of Linear Regression

1 Linearity

The relationship between the independent variable(s) X and the dependent variable Y must be linear. The model is correctly specified. Detection: scatterplot of Y vs X should show a straight-line pattern. If it shows a curve, the linearity assumption is violated.

2 Independence of Errors

The error terms epsilon_i must be independent of each other — there is no correlation between consecutive residuals. This assumption is most important in time-series data. Violation is called autocorrelation. Detection: Durbin-Watson test (DW statistic close to 2 = no autocorrelation).

3 Homoscedasticity (Constant Variance)

The variance of the error terms must be constant across all levels of X: Var(epsilon_i) = sigma^2 for all i. Violation is called Heteroscedasticity. Detection: residual plot — a fan or funnel shape indicates heteroscedasticity. Breusch-Pagan test or White's test can be used formally.

4 **Normality of Errors**

The error terms epsilon_i must be normally distributed: epsilon ~ N(0, sigma^2). This is needed for valid hypothesis tests (t-tests, F-tests) and confidence intervals. In large samples, the Central Limit Theorem reduces the importance of this assumption. Detection: Q-Q plot of residuals; Shapiro-Wilk test.

5 **No Multicollinearity (Multiple Regression)**

In multiple regression, the independent variables must not be highly correlated with each other. Severe multicollinearity makes it impossible to isolate the individual effect of each predictor. Detection: VIF (Variance Inflation Factor) — VIF > 10 indicates serious multicollinearity. Also check the correlation matrix between predictors.

Properties of OLS Estimators — BLUE

The Gauss-Markov Theorem states: Under the 5 assumptions above, OLS estimators are BLUE — Best Linear Unbiased Estimators.

B — Best	OLS estimators have the MINIMUM VARIANCE among all linear unbiased estimators. No other linear unbiased estimator has smaller variance.
L — Linear	OLS estimators are linear functions of the observed values of Y.
U — Unbiased	E(beta_hat_j) = beta_j for all j. On average, the estimator hits the true value.
E — Estimators	They are statistics computed from the sample data to estimate the population parameters.

Additional OLS properties: • Consistency: beta_hat converges to beta as sample size n increases • The OLS line always passes through the point (x-bar, y-bar) • The sum of residuals is always zero: Sum(e_i) = 0 • If any assumption is violated, OLS may no longer be BLUE

Q2 | 10 Marks | 2021-22 Final

Explain the Durbin-Watson test: (i) Define test statistic (ii) Explain the rules for autocorrelation (iii) Assumptions of the Durbin-Watson test

What is Autocorrelation?

Autocorrelation (serial correlation) occurs when the error terms in a regression model are correlated across observations — most commonly across time periods in time-series data. The OLS assumption violated is: Cov(epsilon_i, epsilon_j) = 0 for i ≠ j.

The Durbin-Watson Test Statistic

The Durbin-Watson (DW) statistic tests for first-order autocorrelation — whether the residual at time t is correlated with the residual at time t-1.

DW = Sum[(e_t - e_(t-1))^2] / Sum[e_t^2] for t = 2 to n

e_t = residual at time t | e_(t-1) = residual at the previous time period

Rules for Interpreting the DW Statistic

DW Value	Interpretation	Type of Autocorrelation
Close to 0	Consecutive residuals have the SAME sign — positive pattern	**Strong POSITIVE autocorrelation**
Between 0 and dL	Below lower critical value — reject no-autocorrelation	**Positive autocorrelation**
Between dL and dU	Inconclusive zone — test is not decisive	**Inconclusive**
Close to 2	Residuals are uncorrelated — model is well-specified	**NO autocorrelation (ideal)**
Between 4-dU and 4-dL	Inconclusive zone for negative autocorrelation	**Inconclusive**
Close to 4	Consecutive residuals have OPPOSITE signs — alternating pattern	**Strong NEGATIVE autocorrelation**

Assumptions of the Durbin-Watson Test

- The regression model includes an intercept (constant) term.
- The independent variables X are non-stochastic (fixed in repeated sampling).
- The error terms follow a first-order autoregressive process: epsilon_t = rho * epsilon_(t-1) + u_t
- The model does NOT include a lagged dependent variable as a predictor (e.g. Y_(t-1) as X).
- The observations must be in chronological order — the DW test is specifically for time-series data.
- The sample size n should be large enough (typically n > 15) for the test to be reliable.

> The DW test has a limitation: it only tests for FIRST-ORDER autocorrelation (correlation between adjacent residuals). For higher-order autocorrelation, use the Breusch-Godfrey LM test, which is more general.

Q3 | 5-10 Marks | 2021-22 Final & Re-exam | REPEATED

Explain Heteroscedasticity — its definition, causes, detection methods, and consequences for OLS estimates.

Definition

Heteroscedasticity occurs when the variance of the error terms in a regression model is NOT constant across all levels of the independent variable(s). It violates the OLS assumption of homoscedasticity.

Homoscedasticity (ideal)	Heteroscedasticity (violation)
Var(epsilon_i) = sign *All residuals have th*	**Var(epsilon_i) = sigma_i^2 (varies w** *Residual spread changes with X — bad*
Residual plot: random horizontal band arounc	Residual plot: fan/funnel shape variance grows with fitted values

Common Causes of Heteroscedasticity

Nature of the data

Income data, city populations, and company revenues have wide natural variation at high values — larger firms have more variable costs than smaller ones.

Outliers

Extreme observations in the data can create the appearance of non-constant variance, pulling the residuals unevenly.

Model misspecification

Omitting an important variable or using the wrong functional form can cause the residuals to show a systematic pattern related to the fitted values.

Learning and improvement effects

In longitudinal studies, early measurements are more variable (people or processes are still adapting), while later measurements stabilise.

Data transformation omitted

When the relationship between variables is multiplicative rather than additive, using a linear model without log transformation induces heteroscedasticity.

Detection Methods

1 **Residual Plot (Visual)**
Plot residuals (e_i) against fitted values ($\hat{Y}$) or against each predictor. A fan or funnel shape — where spread increases or decreases — is the classic sign of heteroscedasticity. This is the first and simplest check.

2 **Breusch-Pagan Test (Formal)**
Regresses the squared residuals on the predictors. If the coefficient is significant ($p < 0.05$), heteroscedasticity is present. H0: $Var(\epsilon_i) = \sigma^2$ (homoscedastic). H1: $Var(\epsilon_i)$ varies with X.

3 **White's Test (Formal)**
A more general test that also detects non-linear forms of heteroscedasticity. Does not assume a specific form for the variance function.

4 **Goldfeld-Quandt Test**
Splits the data into two groups (low X and high X) and tests whether the variance in the two groups is significantly different using an F-test.

Consequences for OLS Estimates

Effect 1 — OLS remains UNBIASED: The coefficient estimates $\hat{\beta}$ are still correct on average. Heteroscedasticity does NOT cause bias in OLS coefficients.

Effect 2 — OLS is NO LONGER EFFICIENT: The estimates no longer have minimum variance. Other estimators (like WLS) can produce more precise estimates.

Effect 3 — Standard errors are WRONG: This is the most damaging effect. OLS underestimates standard errors when positive heteroscedasticity is present, making coefficients appear more precise than they are.

Effect 4 — t-statistics and F-statistics are INVALID: Because standard errors are wrong, all hypothesis tests on coefficients are unreliable. Variables may appear significant when they are not (or vice versa).

Effect 5 — R-squared may be MISLEADING: The model may appear to fit well overall but the fit quality varies across the range of X.

Fixes for Heteroscedasticity: 1. Transform the dependent variable: use log(Y) or sqrt(Y) 2. Use Weighted Least Squares (WLS) — give less weight to high-variance observations 3. Use heteroscedasticity-consistent (robust) standard errors (White's HC errors) 4. Re-specify the model — add missing variables or change functional form

Q4 | 5 Marks | 2022-23 Re-exam & 2023-24 Final | REPEATED

What is the difference between R-squared and Adjusted R-squared? When should you prefer Adjusted R-squared?

R-squared (R^2)

R-squared measures the proportion of total variation in Y that is explained by the regression model. It ranges from 0 (model explains nothing) to 1 (model explains everything).

R^2 = SSR / SST = 1 - (SSE / SST)

SSR = Regression Sum of Squares (explained variation) SSE = Error Sum of Squares (unexplained variation) SST = Total Sum of Squares = SSR + SSE

Critical problem with R^2: Adding ANY new variable to the model — even a completely irrelevant one — will NEVER decrease R^2. It will either stay the same or increase. This means R^2 always rewards adding more predictors, regardless of whether they are useful.

Adjusted R-squared (R^2_adj)

Adjusted R-squared penalises the model for adding predictors that do not improve explanatory power. Unlike R^2, it can actually DECREASE when a useless variable is added.

R^2_adj = 1 - [(1 - R^2) * (n - 1) / (n - k - 1)]

n = number of observations | k = number of predictors (not counting the intercept) As k increases with no improvement in fit, the penalty term (n-1)/(n-k-1) grows

Key Differences

Aspect	R-squared (R^2)	Adjusted R-squared
Range	Always 0 to 1	Can be negative (very poor models)
Effect of adding a variable	**NEVER decreases — always increases**	**DECREASES if variable is not useful**

Penalty for complexity	None — no penalty	Yes — penalises extra predictors
Use for model comparison	Unreliable across models with different numbers of predictors	Reliable — accounts for different numbers of predictors
Simple regression (1 predictor)	Appropriate	Same as R^2 approximately

When to prefer Adjusted R^2: • Always use Adjusted R^2 when comparing models with DIFFERENT numbers of predictors • Use it when doing variable selection (deciding which predictors to keep) • Use it in multiple regression — R^2 alone is misleading in multi-predictor models • Rule of thumb: if adding a new variable does NOT increase Adjusted R^2, the variable is not contributing useful information and should be removed

Q5 | 5 Marks | 2023-24 Final

Explain R-squared in Linear Regression — what it measures, how it is calculated, and its limitations.

What R-squared Measures

R-squared (R^2), also called the coefficient of determination, measures the proportion of total variation in the dependent variable Y that is explained by the regression model. It is the most commonly reported measure of model fit.

How R-squared is Calculated

The total variation in Y is decomposed into two parts:

SST (Total)	= Sum[(Yi - Y-bar)^2] *Total variation in Y around its mean — fixed for a given dataset*
SSR (Regression)	= Sum[(Y-hat_i - Y-bar)^2] *Variation explained BY the model (regression line)*
SSE (Error)	= Sum[(Yi - Y-hat_i)^2] *Variation NOT explained by the model (residuals)*
Decomposition	SST = SSR + SSE *Total = Explained + Unexplained*

R^2 = SSR / SST = 1 - (SSE / SST)

Interpretation: R^2 = 0.80 means the model explains 80% of the variation in Y

Limitations of R-squared

Never decreases when adding variables

R^2 always increases (or stays the same) when a new variable is added to the model, even if that variable is completely irrelevant. This makes it easy to artificially inflate R^2 by adding useless predictors.

Does not indicate whether the model is appropriate

A high R^2 does not mean the regression assumptions are met. A non-linear relationship can still produce a high R^2 from a linear model if the range is small.

Cannot compare models with different numbers of predictors

R^2 cannot be used to compare a model with 3 predictors vs a model with 8 predictors — the model with more predictors will almost always have higher R^2 regardless of fit quality.

Does not indicate statistical significance

A high R^2 does not mean the predictors are statistically significant. Always check the F-test and individual t-tests for significance.

Sensitive to outliers

A single outlier can dramatically change R^2 in either direction.

Solution to main limitation: Use ADJUSTED R^2 when comparing models with different numbers of predictors. Adjusted R^2 penalises for adding unnecessary variables.

Q6 | 10 Marks | 2022-23 Re-exam & 2022-23 Final | REPEATED

Briefly explain Multiple Regression. Mention assumptions and how to interpret coefficients in a multiple regression output.

What is Multiple Regression?

Multiple regression extends simple linear regression by using TWO OR MORE independent variables (predictors) to explain and predict the dependent variable Y. It allows us to control for confounding variables and isolate the unique effect of each predictor.

Y = beta_0 + beta_1*X1 + beta_2*X2 + ... + beta_k*Xk + epsilon

Y = dependent variable | X1, X2, ..., Xk = k independent variables (predictors) beta_0 = intercept | beta_j = partial regression coefficient for Xj epsilon = error term | k = number of predictors

Assumptions of Multiple Regression

All 5 assumptions from Q1 apply — plus the additional assumption:

1. **Linearity**
The relationship between each predictor Xj and Y is linear, holding all other predictors constant.

2. **Independence of Errors**
No autocorrelation in residuals — Cov(epsilon_i, epsilon_j) = 0.

3. **Homoscedasticity**
Var(epsilon_i) = sigma^2 is constant — no heteroscedasticity.

4. **Normality of Errors**
epsilon ~ N(0, sigma^2) for valid inference.

5. **No Multicollinearity**
The predictor variables X1, X2, ..., Xk must NOT be highly correlated with each other. If they are, it becomes impossible to separate their individual effects on Y. VIF > 10 indicates serious multicollinearity.

How to Interpret Coefficients in Regression Output

Each coefficient beta_j in a multiple regression has a specific interpretation:

Coefficient	Interpretation
beta_0 (intercept)	The expected value of Y when ALL predictors X1, X2, ..., Xk equal zero. May not have meaningful interpretation depending on context.
beta_j (slope for Xj)	For a one-unit increase in Xj, Y changes by beta_j units, HOLDING ALL OTHER PREDICTORS CONSTANT. This 'ceteris paribus' (all else equal) interpretation is what makes multiple regression powerful.
Positive beta_j	Xj has a POSITIVE relationship with Y — Y increases as Xj increases.
Negative beta_j	Xj has a NEGATIVE relationship with Y — Y decreases as Xj increases.
p-value for beta_j	Tests H0: beta_j = 0 (Xj has no effect on Y). If $p < 0.05$, the predictor is statistically significant.

Example interpretation: In a model predicting Salary (Y) using Years_Experience (X1) and Education_Level (X2): beta_1 = 5000 means: each additional year of experience increases salary by Rs 5,000, HOLDING education level constant. beta_2 = 20000 means: each additional level of education increases salary by Rs 20,000, HOLDING years of experience constant.

Q7 | 10 Marks | 2022-23 Final

State the full list of assumptions of linear regression: Linearity, Independence of errors, Homoscedasticity, Normality of errors, No Multicollinearity.

This question asks for the same 5 assumptions as Q1. For a 10-mark answer, write each assumption with: (1) the formal statement, (2) what happens if it is violated, and (3) how to detect the violation. See Q1 for the full detailed answers.

Complete Assumptions — Quick Reference

Assumption	Formal Statement	Violation	Detection
1. Linearity	E(Y|X) = beta_0 + beta_1*X	Non-linearity / model misspecification	Residual vs fitted plot (curved pattern)
2. Independence of Errors	Cov(e_i, e_j) = 0 for i != j	Autocorrelation / serial correlation	Durbin-Watson test (DW near 2 = OK)
3. Homoscedasticity	Var(e_i) = sigma^2 (constant)	Heteroscedasticity (fan-shaped residuals)	Residual plot; Breusch-Pagan test
4. Normality of Errors	epsilon ~ N(0, sigma^2)	Non-normal residuals (skewed distribution)	Q-Q plot; Shapiro-Wilk test
5. No Multicollinearity	Predictors are not perfectly correlated	Inflated SEs; unstable coefficients	VIF > 10; correlation matrix

Remember: Under all 5 assumptions, OLS is BLUE (Best Linear Unbiased Estimator). Violating any assumption weakens the guarantees. The Gauss-Markov theorem is the theoretical basis for OLS being optimal.

Q8 | 5 Marks | 2022-23 Final

Explain Potential Problems of Regression: multicollinearity, outliers, non-linearity, heteroscedasticity, autocorrelation.

5 Potential Problems of Regression

Multicollinearity

Definition	Two or more predictors in a multiple regression are highly correlated with each other.
Consequence	Coefficients become very sensitive to small changes in data. Standard errors are inflated, making t-statistics small and predictors appear insignificant even when the overall model fits well. The individual effect of each predictor cannot be reliably estimated.
Detection	VIF > 10 indicates serious multicollinearity. Correlation matrix between predictors.
Fix	Drop one of the correlated predictors. Combine them into an index. Use Ridge regression (L2 penalty).

Outliers

Definition	Data points that lie far from the fitted regression line — extreme values in Y.
Consequence	Outliers have excessive influence on the OLS estimates. A single outlier can dramatically change the slope and intercept. They inflate SSE and reduce R^2.
Detection	Cook's Distance > 1 indicates high influence. Studentised residual plot.
Fix	Investigate cause of outlier. If data error — fix it. If genuine — consider robust regression.

Non-linearity

Definition	The relationship between X and Y is curved rather than straight — the linearity assumption fails.
Consequence	Coefficients are biased. The model systematically over-predicts in some regions and under-predicts in others. Residuals show a clear U-shape or curve.
Detection	Residual vs fitted plot shows a systematic curve. Scatterplot of Y vs X shows a curve.

Fix	Apply a transformation (log X, X^2). Use polynomial regression. Use non-linear regression.

Heteroscedasticity

Definition	The variance of the error terms is not constant — it changes with the level of X.
Consequence	OLS remains unbiased but standard errors are WRONG. This makes all hypothesis tests on coefficients (t-tests, F-tests) unreliable. Confidence intervals are incorrect.
Detection	Residual plot shows fan/funnel shape. Breusch-Pagan test. White's test.
Fix	Transform Y (log Y). Use Weighted Least Squares. Use robust (White's HC) standard errors.

Autocorrelation

Definition	The error terms are correlated across observations — most common in time-series data.
Consequence	OLS remains unbiased but is no longer efficient. Standard errors are underestimated, inflating t-statistics. Hypothesis tests are unreliable. R^2 may be overestimated.
Detection	Durbin-Watson test (DW close to 2 = no autocorrelation). Residual plot vs time.
Fix	Add omitted variables. Use GLS / Cochrane-Orcutt procedure. Include lagged variables.

Q9 | 5 Marks | 2023-24 Re-exam

Describe the use of dummy variables or one-hot encoding variables in regression models. Explain with an example.

What are Dummy Variables?

Linear regression requires all variables to be numeric. However, many real-world predictors are categorical (qualitative) — like gender, region, or season. Dummy variables (also called indicator variables) are binary (0 or 1) variables created to represent categorical predictors numerically in a regression model.

Dummy variable D_i = 1 if observation belongs to category i Dummy variable D_i = 0 if observation does NOT belong to category i

Each category becomes a binary 0/1 column that can be used in regression

The Dummy Variable Trap — Dummy Variable Rule

CRITICAL RULE: For a categorical variable with k categories, you must include EXACTLY k-1 dummy variables in the model (not k). The excluded category is called the BASE CATEGORY or REFERENCE CATEGORY. All coefficients are interpreted relative to this base. Including all k dummies causes PERFECT MULTICOLLINEARITY (the dummy variable trap) — the dummies sum to 1 which equals the intercept column, making OLS impossible.

Example — Gender in a Salary Regression

Categorical variable: Gender with 2 categories — Male, Female Number of dummies needed: k - 1 = 2 - 1 = 1 dummy variable Base category: Male (assigned 0) Dummy variable: D = 1 if Female, D = 0 if Male

Salary = beta_0 + beta_1 * Years_Experience + beta_2 * D_Female + epsilon

D_Female = 1 if the employee is Female, 0 if Male

Interpretation of coefficients:

beta_0	Expected salary for a MALE employee with 0 years experience *Intercept for the base category (Male)*
beta_1	Change in salary per additional year of experience, for BOTH genders *Same slope assumed for both groups*
beta_2	Additional salary for a FEMALE employee compared to a MALE employee, holding years of experience constant *If beta_2 < 0: female employees earn less on average*

Example — Season (4 Categories) in Sales Regression

Categorical variable: Season with 4 categories — Spring, Summer, Autumn, Winter Number of dummies needed: k - 1 = 4 - 1 = 3 dummy variables Base category: Spring (excluded — assigned 0 for all three dummies) Dummies: D_Summer, D_Autumn, D_Winter

Sales = beta_0 + beta_1*D_Summer + beta_2*D_Autumn + beta_3*D_Winter + epsilon

Spring is the reference — its effect is captured in the intercept beta_0

Interpretation: beta_1 = change in sales in Summer COMPARED TO Spring. If beta_1 = 500, then Summer sales are Rs 500 higher than Spring sales on average. To get Winter vs Summer, compute beta_3 - beta_1.

Q10 | 10 Marks | 2023-24 Re-exam

How are qualitative predictors incorporated into the regression model? Explain with an example. (Dummy variable coding, interpretation of coefficients)

Why We Need Qualitative Predictors in Regression

Many important predictors in social science, business, and economics are qualitative (categorical) — gender, geographic region, industry type, education level, marital status. Since OLS regression requires numerical inputs, we must ENCODE categorical variables as numbers before including them. The standard method is dummy variable coding.

Step-by-Step: How to Incorporate a Qualitative Predictor

1 **Identify the qualitative variable and its categories**

Determine the variable (e.g. Region: North, South, East, West) and count the categories k=4.

2 **Choose a base (reference) category**

Select one category to be the reference group. All other categories will be compared to this. Choice of reference category does not change model fit — only the interpretation of coefficients. Usually choose the most common or most natural reference group.

3 **Create k-1 dummy variables**

Create one binary (0/1) dummy variable for each category EXCEPT the base. For Region with k=4 and base=North: create D_South, D_East, D_West. Each observation gets a 1 in exactly one dummy column (or 0 in all if it is the base).

4 **Include the dummy variables in the regression**

Add D_South, D_East, D_West as predictors alongside any other numeric predictors. Do NOT include all k dummies — this causes the dummy variable trap (perfect multicollinearity).

5 **Interpret each dummy coefficient**

Each dummy coefficient = the DIFFERENCE in the mean of Y between that category and the base category, holding all other predictors constant.

Full Worked Example — House Price Prediction

Dataset: House prices with predictors — Size (sq ft), Bedrooms, and Location type Location types: Urban, Suburban, Rural (k = 3 categories) Base category chosen: Rural Dummies created: D_Urban and D_Suburban (k-1 = 2 dummies)

Price = beta_0 + beta_1*Size + beta_2*Bedrooms + beta_3*D_Urban + beta_4*D_Suburban + epsilon

Base category = Rural | D_Urban = 1 if Urban, 0 otherwise | D_Suburban = 1 if Suburban, 0 otherwise

Suppose OLS gives:

Coefficient	Estimated Value	Interpretation
beta_0 (intercept)	10,00,000	Base price of a RURAL house with 0 size and 0 bedrooms
beta_1 (Size)	2,000	Each additional sq ft adds Rs 2,000 to price, all else equal
beta_2 (Bedrooms)	50,000	Each additional bedroom adds Rs 50,000, all else equal
beta_3 (D_Urban)	3,00,000	Urban houses cost Rs 3,00,000 MORE than Rural houses (same size & bedrooms)
beta_4 (D_Suburban)	1,50,000	Suburban houses cost Rs 1,50,000 MORE than Rural houses (same size & bedrooms)

Urban vs Suburban comparison: beta_3 - beta_4 = 3,00,000 - 1,50,000 = 1,50,000. Urban houses cost Rs 1,50,000 more than Suburban houses (holding size and bedrooms constant). Note: The reference category (Rural) does not have its own dummy variable — its effect is fully captured in the intercept beta_0.

Key rule summary: • k categories --> k-1 dummy variables (never k, to avoid the dummy variable trap) • All dummy coefficients are interpreted as differences FROM the base category • The choice of base category does not affect model fit (R^2, F-statistic) — only the coefficient values and their interpretation change • This exact approach is also called ONE-HOT ENCODING in machine learning